

Reversible Reactions

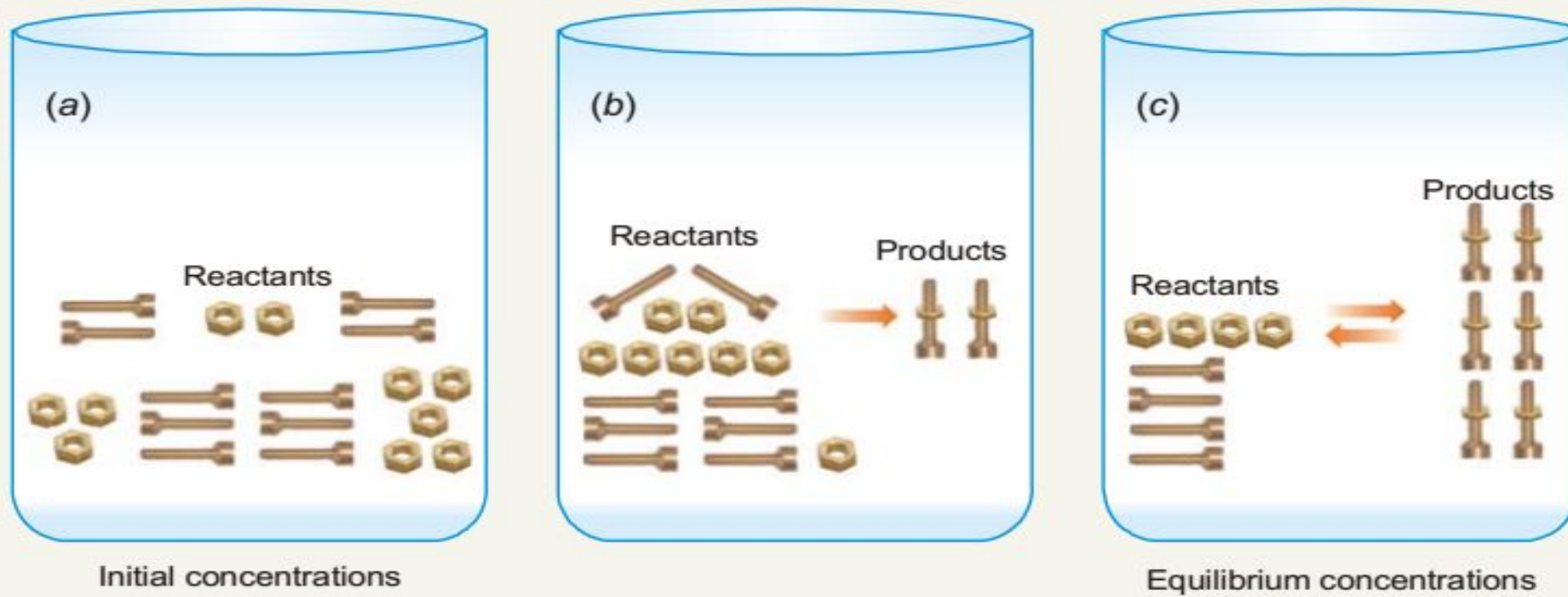
Not all chemical reactions proceed to completion. In most reactions two or more substances react to form products which themselves react to give back the original substances. Thus A and B may react to form C and D which react together to reform A and B.

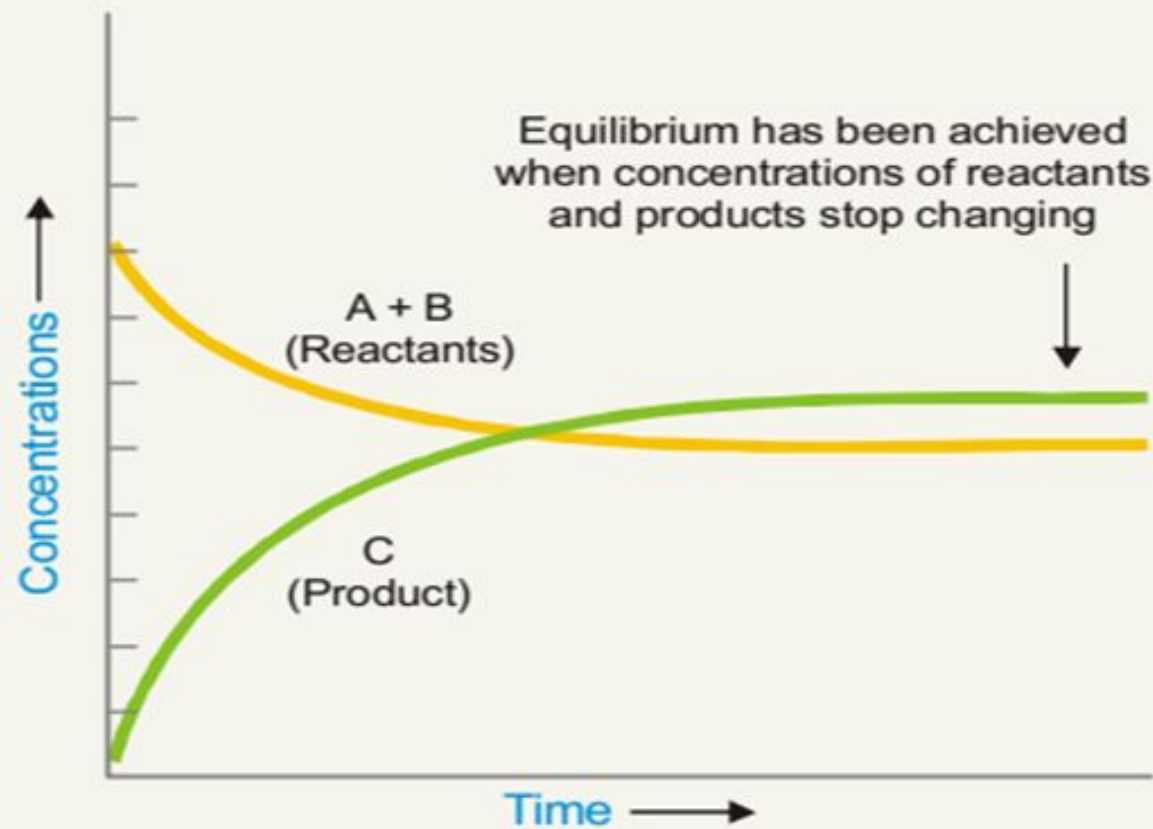
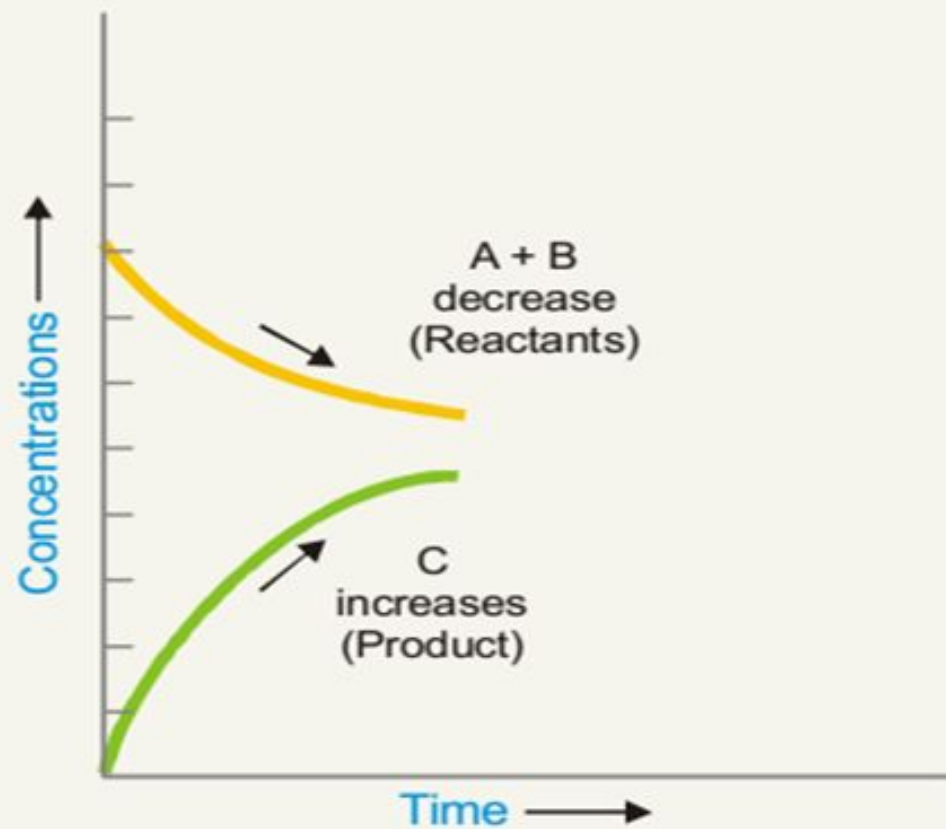


A reaction which can go in the forward and backward direction simultaneously is called a Reversible reaction. Such a reaction is represented by writing a pair of arrows between the reactants and products.



Nuts and Bolts Representation of Chemical Equilibrium

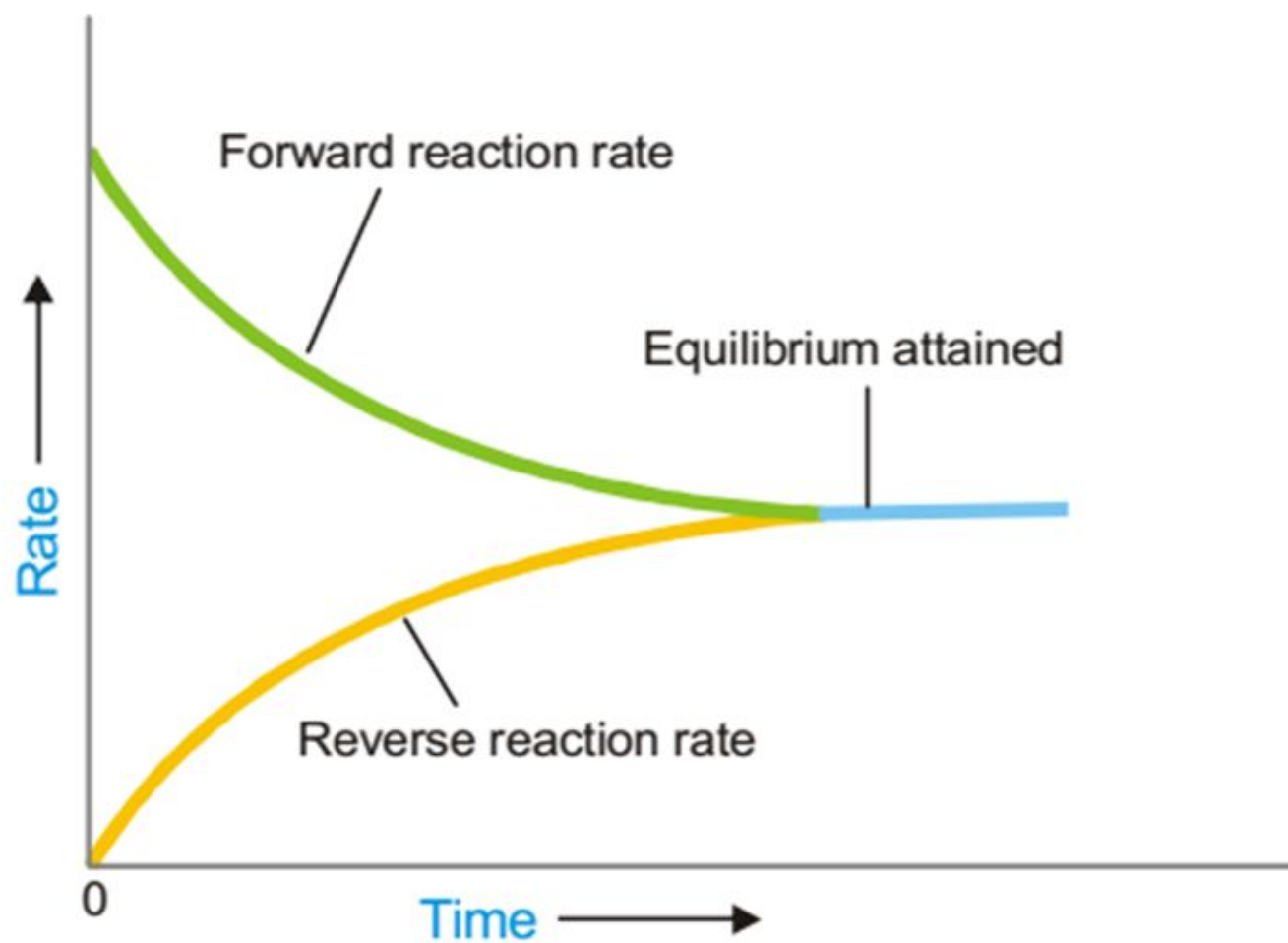




Let us consider the reaction



If we start with A and B in a closed vessel, the forward reaction proceeds to form C and D. The concentrations of A and B decrease and those of C and D increase continuously. As a result the rate of forward reaction also decreases and the rate of the reverse reaction increases. Eventually, the rate of the two opposing reactions equals and the system attains a state of equilibrium. Thus Chemical equilibrium may be defined as: the state of a reversible reaction when the two opposing reactions occur at the same rate and the concentrations of reactants and products do not change with time. Furthermore, the true equilibrium of a reaction can be attained from both sides. Thus the equilibrium concentrations of the reactants and products are the same whether we start with A and B, or C and D.

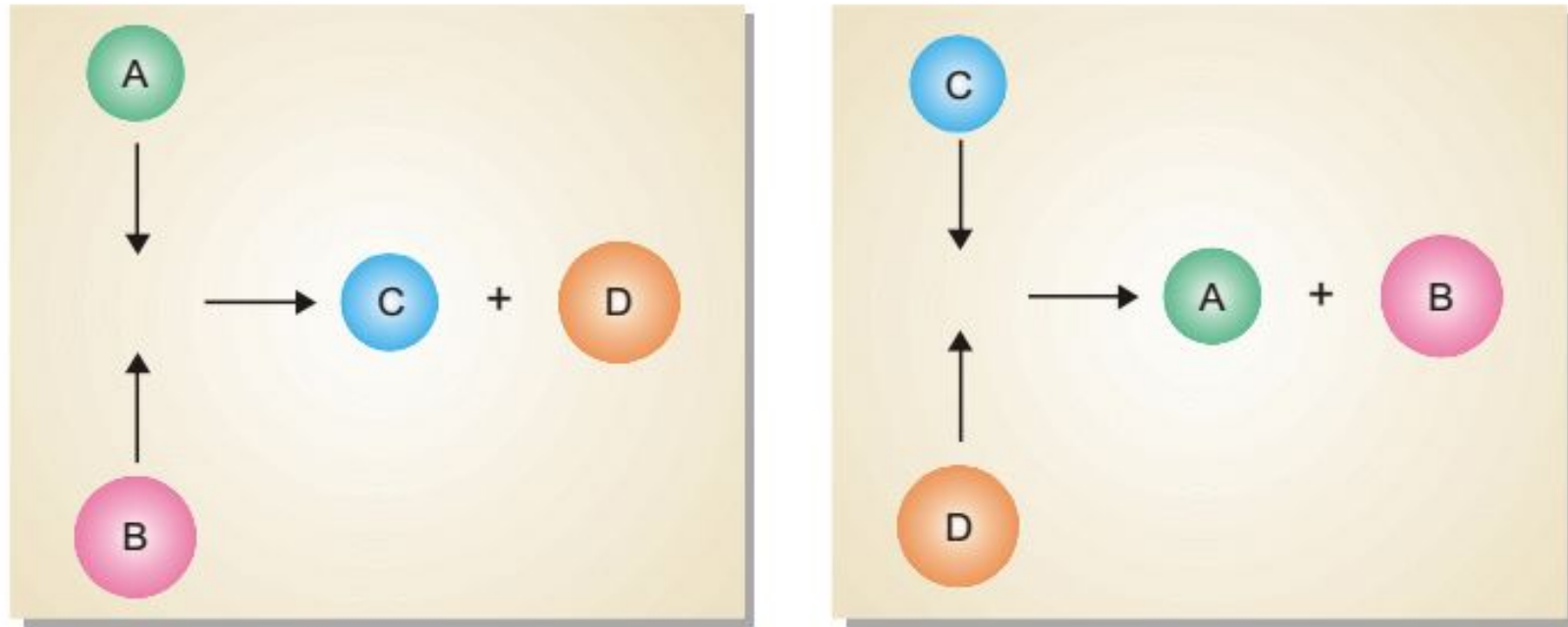


■ **Figure 17.1**

At equilibrium the forward reaction rate equals the reverse reaction rate.

Chemical Equilibrium is Dynamic Equilibrium

We have shown above that as the reaction $A + B = C + D$ attains equilibrium, the concentrations of A and B, as also of C and D remain constant with time. Apparently it appears that the equilibrium is dead. But it is not so. The equilibrium is dynamic. Actually, the forward and the reverse reactions are taking place at equilibrium but the concentrations remain unchanged.



■ **Figure 17.2**

Molecules of A and B colliding to give C and D, and those of C and D colliding to give A and B.

The dynamic nature of chemical equilibrium can be easily understood on the basis of the kinetic molecular model. The molecules of A and B in the equilibrium mixture collide with each other to form C and D. Likewise C and D collide to give back A and B. The collisions of molecules in a closed system is a ceaseless phenomenon. Therefore collisions of A and B giving C and D (Forward reaction) and collisions of C and D giving back A and B (reverse reaction) continue to occur even at equilibrium, while concentrations remain unchanged.

Characteristics Of Chemical Equilibrium

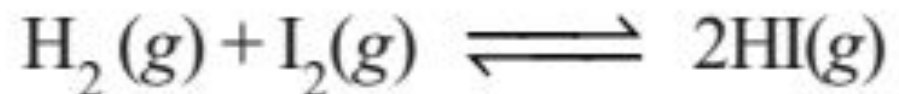
1. Constancy of concentrations

When a chemical equilibrium is established in a closed vessel at constant temperature, concentrations of the various species in the reaction mixture become constant. The reaction mixture at equilibrium is called **Equilibrium mixture**.

The concentrations at equilibrium are called Equilibrium concentrations. The equilibrium concentrations are represented by square brackets with subscript eq or eqm, $[]_{\text{eq}}$. Thus $[A]_{\text{eq}}$ denotes the equilibrium concentration of substance A in moles per litre. In modern practice the subscript eq is not used.

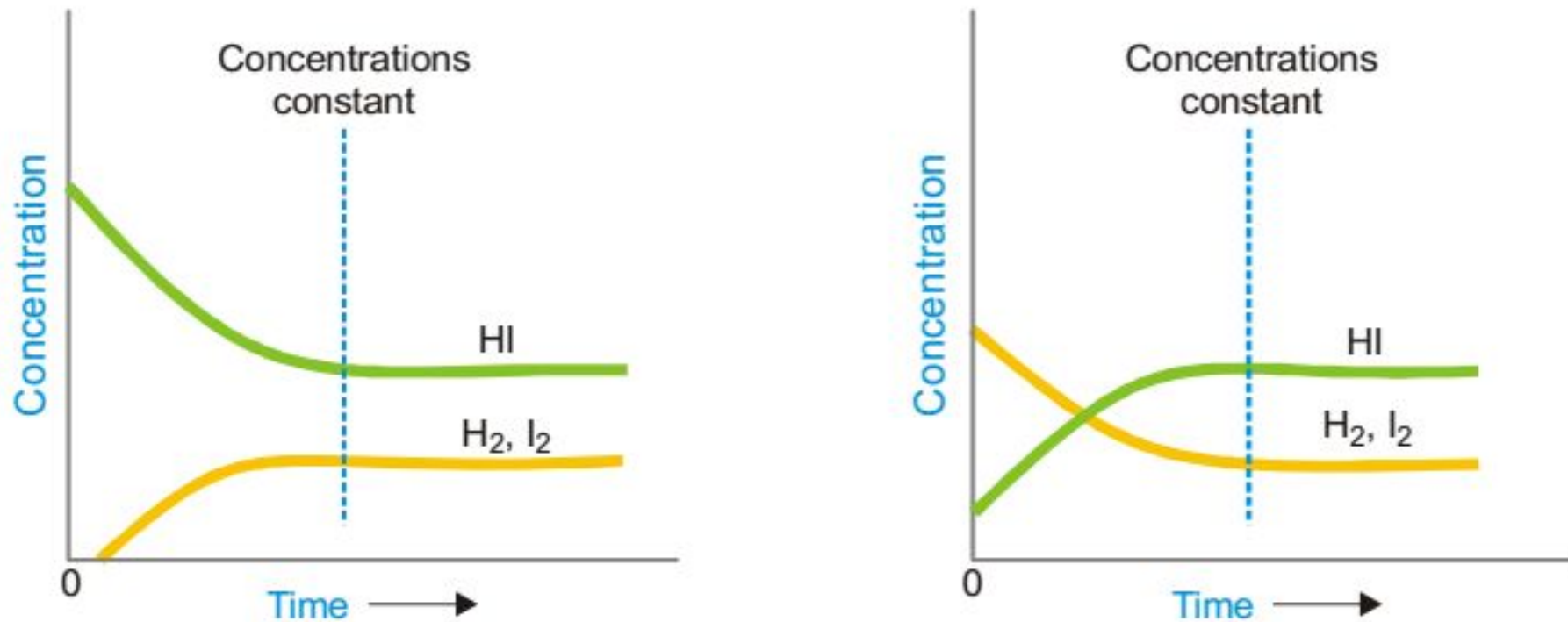
2. Equilibrium can be initiated from either side

The state of equilibrium of a reversible reaction can be approached whether we start with reactants or products, for example, the equilibrium



is established if we start the reaction with H_2 and I_2 , or 2 HI.

Fig. 17.3. Shows whether we start with 1 mole of I_2 and 1 mole of H_2 , or with 2 moles of HI in vessel of the same volume, the same mixture of H_2 , I_2 and HI is obtained.

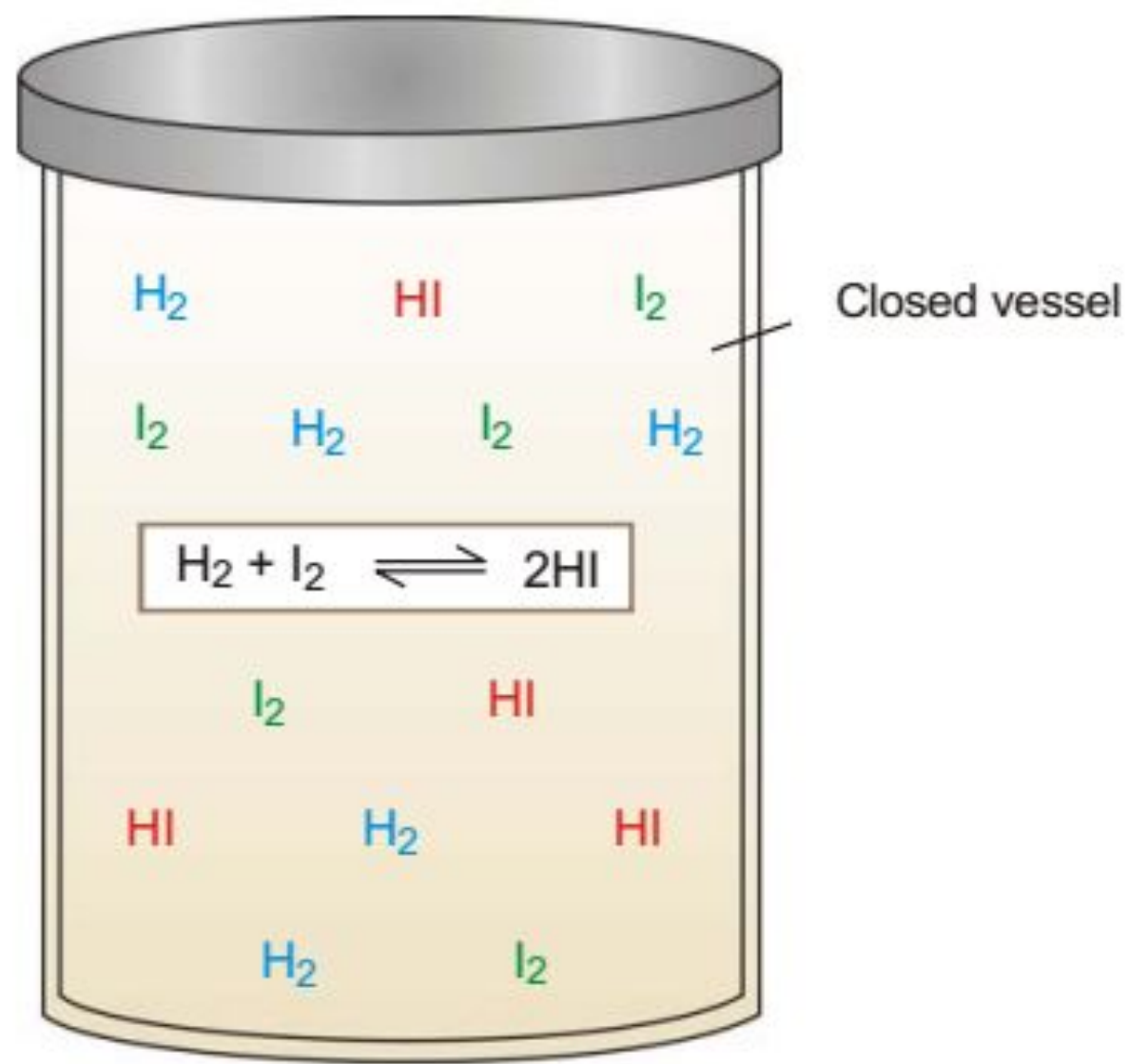


■ **Figure 17.3**

The left graph shows how equilibrium is attained for start with H_2 and I_2 . The right graph depicts the attainment of equilibrium for start with 2HI . Equilibrium concentrations in both cases are the same.

(3) Equilibrium cannot be Attained in an Open Vessel

The equilibrium can be established only if the reaction vessel is closed and no part of the reactants or products is allowed to escape out. In an open vessel, the gaseous reactants and/or products may escape into the atmosphere leaving behind no possibility of attaining equilibrium.

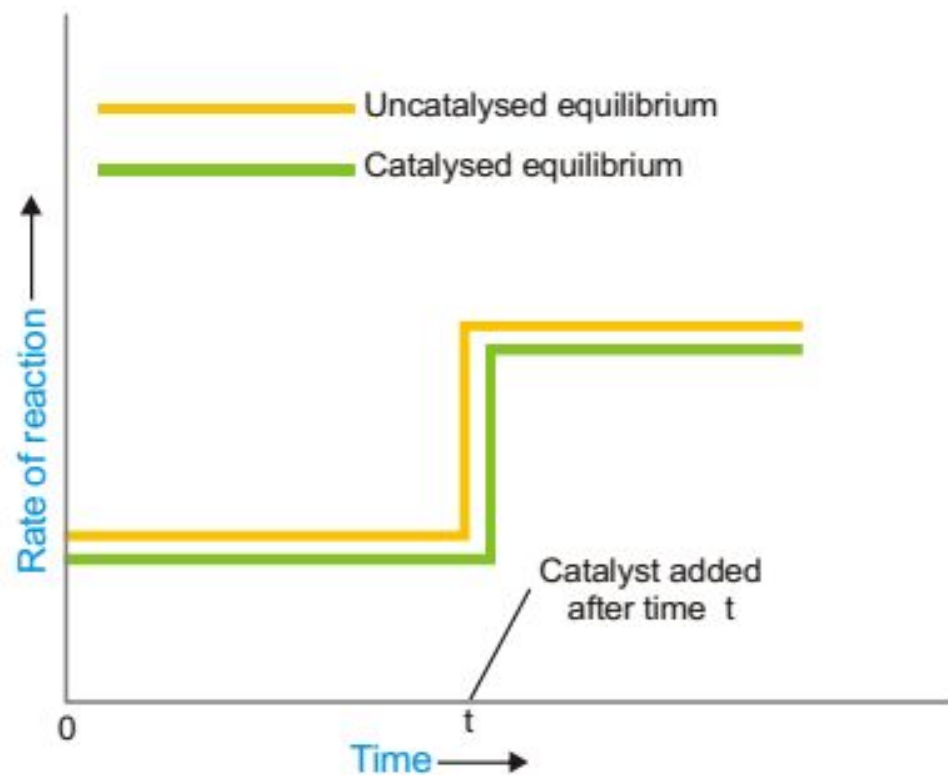


■ **Figure 17.4**

A chemical equilibrium between H_2 , I_2 and HI .

(4) A catalyst cannot change the equilibrium point

When a catalyst is added to a system in equilibrium, it speeds up the rate of both the forward and the reverse reaction to an equal extent. Therefore a catalyst cannot change the equilibrium point except that it is achieved earlier. This enhances the rate of the reaction. The rapid increase of the rate of an equilibrium reaction on the addition of a catalyst is illustrated in Fig. 17.5.



■ **Figure 17.5**

The vertical part of the curve indicates rapid increase of rate of reaction on addition of catalyst.

(5) Value of Equilibrium Constant does not depend upon the initial concentration of reactants

It has been found that equilibrium constant must be the same when the concentrations of reacting species are varied over a wide range.

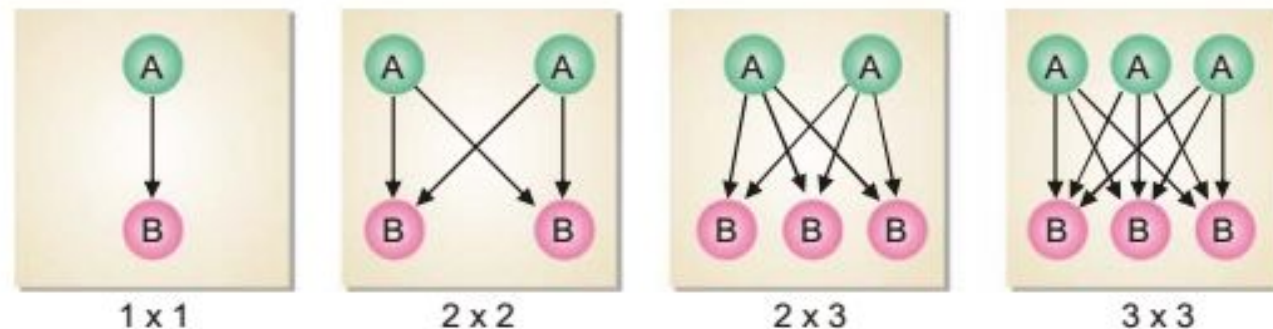
6) At Equilibrium $\Delta G = 0$

At equilibrium the Gibbs free energy (G) is minimum and any change taking place at equilibrium proceeds without change in free energy i.e. $\Delta G = 0$.

LAW OF MASS ACTION

Two Norwegian chemists, Guldberg and Waage, studied experimentally a large number of equilibrium reactions. In 1864, they postulated a generalization called the Law of Mass action.

It states that : the rate of a chemical reaction is proportional to the active masses of the reactants. By the term ‘**active mass**’ is meant the molar concentration i.e., number of moles per litre. It is expressed by enclosing the formula of the substance in square brackets. Thus a gas mixture containing 1 g of hydrogen (H_2) and 127 g of iodine (I_2) per litre has the concentrations $[H_2] = 0.5$ and $[I_2] = 0.5$

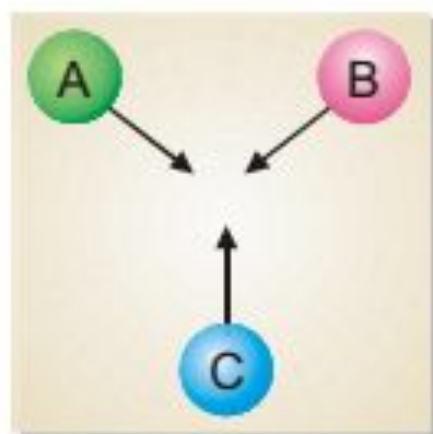


■ **Figure 17.6**

Possible collisions between molecules of the reactants A and B.

Thus in general, we can say that the possibilities of collisions between the reacting molecules A and B are equal to the product of the number of molecules of each species per unit volume. Since the rate of reaction is determined by molecular impacts, it is proportional to moles per unit volume i.e., molar concentration. Thus we can write

$$\begin{aligned}\text{Rate of reaction} &\propto [A][B] \\ &= k[A][B]\end{aligned}$$



■ Figure 17.7

In a reaction, $A + B + C \longrightarrow$ the number of collisions between A, B and C depends on the concentration of each reactant (Fig. 17.7). Therefore,

$$\text{Rate of reaction} \propto [A][B][C] = k[A][B][C]$$

From the above considerations, it stands proved that the rate of a reaction is proportional to the molar concentrations of the reactants.

Equilibrium Constant : Equilibrium Law

Let us consider a general reaction



and let $[A]$, $[B]$, $[C]$ and $[D]$ represent the molar concentrations of A, B, C and D at the equilibrium point. According to the Law of Mass action.

$$\text{Rate of forward reaction} \propto [A][B] = k_1 [A][B]$$

$$\text{Rate of reverse reaction} \propto [C][D] = k_2 [C][D]$$

where k_1 and k_2 are rate constants for the forward and reverse reactions.

At equilibrium, rate of forward reaction = rate of reverse reaction.

Therefore,

$$k_1 [A][B] = k_2 [C][D]$$

or
$$\frac{k_1}{k_2} = \frac{[C][D]}{[A][B]}$$

At any specific temperature k_1/k_2 is constant since both k_1 and k_2 are constant. k_1/k_2 is called **Equilibrium constant** and is represented by the symbol K_c . 'c' indicates that the value is in terms of concentrations of reactants and may be written as

Here, the forward reaction is dependent on the collisions of each of two A molecules. Therefore, for writing the equilibrium expression, each molecule is regarded as a separate entity *i.e.*,



Then the equilibrium constant expression is

$$K_c = \frac{[C][D]}{[A][A]} = \frac{[C][D]}{[A]^2} \text{--- Power equal to coefficient of A}$$

As a general rule, if there are two or more molecules of the same substance in the chemical equation, **its concentration is raised to the power equal to the numerical coefficient of the substance in the equation.**

Equilibrium Constant Expression for a Reaction in General Terms

The general reaction may be written as

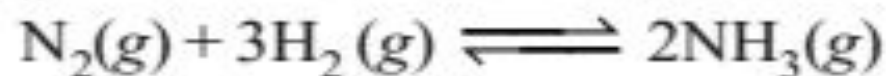


where a , b , c and d are numerical quotients of the substance, A, B, C and D respectively. The equilibrium constant expression is

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

where K_c is the **Equilibrium constant**. The general definition of the equilibrium constant may thus be stated as : **the product of the equilibrium concentrations of the products divided by the product of the equilibrium concentrations of the reactants, with each concentration term raised to a power equal to the coefficient of the substance in the balanced equation.**

Give the equilibrium constant expression for the reaction

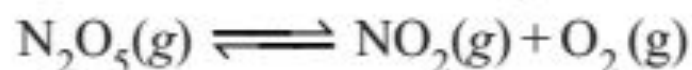


SOLUTION

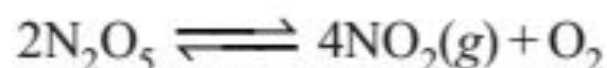
- (1) The equation is already balanced. The numerical quotient of H_2 is 3 and of NH_3 is 2.
- (2) The concentration of the 'product' NH_3 is $[\text{NH}_3]^2$
- (3) The product of concentrations of the reactants is $[\text{N}_2][\text{H}_2]^3$
- (4) Therefore, the equilibrium constant expression is

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$$

SOLVED PROBLEM 2. Write the equilibrium constant expression for the reaction



- (1) The equation as written is not balanced. Balancing yields



- (2) The quotient of the product NO_2 is 4 and of the reactant N_2O_5 it is 2.

- (3) The product of the concentrations of products is

$$[\text{NO}_2]^4 [\text{O}_2]$$

- (4) The concentration of the reactant is

$$[\text{N}_2\text{O}_5]^2$$

- (5) The equilibrium constant expression can be written as

$$K_c = \frac{[\text{NO}_2]^4 [\text{O}_2]}{[\text{N}_2\text{O}_5]^2}$$

EQUILIBRIUM CONSTANT EXPRESSION IN TERMS OF PARTIAL PRESSURES

When all the reactants and products are gases, we can also formulate the equilibrium constant expression in terms of partial pressure. The relationship between the partial pressure (p) of any one gas in the equilibrium mixture and the molar concentration follows from the general ideal gas equation

$$pV = nRT \quad \text{or} \quad p = \left(\frac{n}{V} \right) RT$$

The quantity $\frac{n}{V}$ is the number of moles of the gas per unit volume and is simply the molar concentration. Thus,

$$p = (\text{Molar concentration}) \times RT$$

i.e., the partial pressure of a gas in the equilibrium mixture is directly proportional to its molar concentration at a given temperature. Therefore, we can write the equilibrium constant expression in terms of partial pressure instead of molar concentrations. For a general reaction

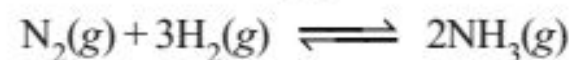


the equilibrium law or the equilibrium constant may be written as

$$K_p = \frac{(p_Y)^y (p_Z)^z}{(p_L)^l (p_M)^m}$$

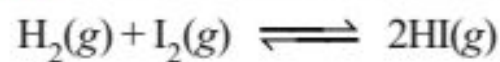
Here K_p is the equilibrium constant, the subscript p referring to partial pressure. Partial pressures are expressed in atmospheres.

SOLVED PROBLEM 1. Write the equilibrium constant expression for the synthesis of ammonia,



$$K_p = \frac{(p_{\text{NH}_3})^2}{(p_{\text{N}_2})(p_{\text{H}_2})^3}$$

SOLVED PROBLEM 2. Using partial pressures, write the equilibrium law for the reaction



$$K_p = \frac{(p_{\text{HI}})^2}{(p_{\text{H}_2})(p_{\text{I}_2})}$$

SOLVED PROBLEM 3. What is the expression K_p for the reaction



For K_p we use the partial pressure in the equilibrium constant expression. Therefore,

$$K_p = \frac{(p_{\text{NO}_2})^2}{(p_{\text{N}_2\text{O}_4})}$$

HOW K_c AND K_p ARE RELATED?

Let us consider a general reaction



where all reactants and products are gases. We can write the equilibrium constant expression in terms of partial pressures as

$$K_p = \frac{(p_C)^l (p_D)^m}{(p_A)^j (p_B)^k} \quad \dots(1)$$

Assuming that all these gases constituting the equilibrium mixture obey the ideal gas equation, the partial pressure (p) of a gas is

$$p = \left(\frac{n}{V} \right) RT$$

Where $\frac{n}{V}$ is the molar concentration. Thus the partial pressures of individual gases, A, B, C and D are:

$$P_A = [A] RT; p_B = [B] RT; p_C = [C] RT; p_D = [D] RT$$

Substituting these values in equation (1), we have

$$K_p = \frac{[C]^l (RT)^l [D]^m (RT)^m}{[A]^j (RT)^j [B]^k (RT)^k}$$

or

$$K_p = \frac{[C]^l [D]^m}{[A]^j [B]^k} \times \frac{(RT)^{l+m}}{(RT)^{j+k}}$$

$$K_p = K_c \times (RT)^{(l+m)-(j+k)}$$

$$\therefore K_p = K_c \times (RT)^{\Delta n} \quad \dots(2)$$

where $\Delta n = (l + m) - (j + k)$, the difference in the sums of the coefficients for the gaseous products and reactants.

From the expression (2) it is clear that when $\Delta n = 0$, $K_p = K_c$.

SOLVED PROBLEM 1. At 500°C, the reaction between N₂ and H₂ to form ammonia has $K_c = 6.0 \times 10^{-2}$. What is the numerical value of K_p for the reaction?

SOLUTION

Here, we will use the general expression

$$K_p = K_c (RT)^{\Delta n}$$

For the reaction $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$

we have $\Delta n = (\text{sum of quotients of products}) - (\text{sum of quotients of reactants})$

$$= 2 - 4 = -2$$

$$K_c = 6.0 \times 10^{-2}$$

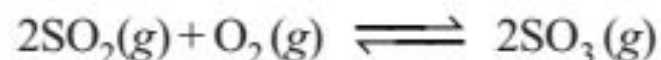
$$T = 500 + 273 = 773 \text{ K}$$

$$R = 0.0821$$

Substituting the value of R , T , K_c and Δn in the general expression, we have

$$\begin{aligned} K_p &= (6.0 \times 10^{-2}) [(0.0821) \times (773)]^{-2} \\ &= \mathbf{1.5 \times 10^{-5}} \end{aligned}$$

SOLVED PROBLEM 1. At equilibrium for the reaction



the concentrations of reactants and products at 727°C were found to be : $\text{SO}_2 = 0.27 \text{ mol L}^{-1}$; $\text{O}_2 = 0.40 \text{ mol L}^{-1}$; and $\text{SO}_3 = 0.33 \text{ mol L}^{-1}$. What is the value of the equilibrium constant, K_c , at this temperature?

SOLUTION

Write the equilibrium constant expression for the reaction

$$K_c = \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2 [\text{O}_2]}$$

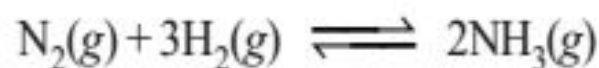
We know that:

$$[\text{SO}_3] = 0.33 \text{ mol L}^{-1}; [\text{SO}_2] = 0.27 \text{ mol L}^{-1}; [\text{O}_2] = 0.40 \text{ mol L}^{-1}$$

Substituting these values in the equilibrium constant expression, we have

$$K_c = \frac{(0.33)^2}{(0.27)^2 \times (0.40)} = 3.7 \text{ mol l}^{-1} \text{ or simply } 3.7$$

SOLVED PROBLEM 2. Some nitrogen and hydrogen gases are pumped into an empty five-litre glass bulb at 500°C. When equilibrium is established, 3.00 moles of N₂, 2.10 moles of H₂ and 0.298 mole of NH₃ are found to be present. Find the value of K_c for the reaction



at 500°C.

SOLUTION

The equilibrium concentrations are obtained by dividing the number of moles of each reactant and product by the volume, 5.00 litres. Thus,

$$[\text{N}_2] = 3.00 \text{ mole}/5.00 \text{ L} = 0.600 \text{ M}$$

$$[\text{H}_2] = 2.10 \text{ mole}/5.00 \text{ L} = 0.420 \text{ M}$$

$$[\text{NH}_3] = 0.298 \text{ mole}/5.00 \text{ L} = 0.0596 \text{ M}$$

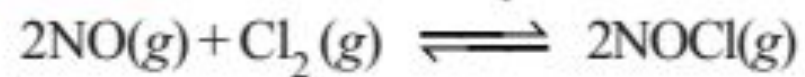
Substituting these concentrations (not number of moles) in the equilibrium constant expression, we get the value of K_c .

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3} = \frac{(0.0596)^2}{(0.600)(0.420)^3} = 0.080$$

Thus, for the reaction of H₂ and N₂ to form NH₃ at 500°C, we can write

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3} = \mathbf{0.080}$$

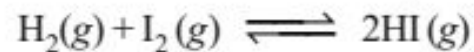
SOLVED PROBLEM 2. The value of K_p at 25°C for the reaction



is $1.9 \times 10^3 \text{ atm}^{-1}$. Calculate the value of K_c at the same temperature.

SOLUTION:

SOLVED PROBLEM 5. 13.5 ml of HI are produced by the interaction of 8.1 ml of hydrogen and 9.3 ml of iodine vapour at 444°C. Calculate the equilibrium constant at this temperature of the reaction.



SOLUTION

The equilibrium constant expression for the reaction is

$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$$

Let us proceed to find the equilibrium concentrations.

Since the number of moles in a gas under the same conditions of temperature and pressure are proportional to volumes, ml of a gas may be used instead of molar concentrations. Thus,



Equilibrium volumes : (8.1-x) (9.3-x) 2x

The experimental volume of HI = 13.5 ml (given)

$$\therefore 2x = 13.5 \text{ ml}$$

$$\text{or } x = 6.75 \text{ ml}$$

Thus, the equilibrium concentrations are:

$$[\text{H}_2] = (8.1 - x) = 8.1 - 6.75 = 1.35$$

$$[\text{I}_2] = (9.3 - x) = 9.3 - 6.75 = 2.55$$

$$[\text{HI}] = 13.5$$

Substituting these values in the equilibrium constant expression, we have

$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} = \frac{(13.5)^2}{1.35 \times 2.55} = \mathbf{52.94}$$